

Instructions:

1. This question paper contains 50 questions. All questions are compulsory.
2. The duration of the examination is 90 minutes.
3. Each question carries 2 marks. There are no negative marks for wrong answers.
4. Use only blue or black ball-point pen to mark your answers on the OMR Answer Sheet.
5. The use of calculators, mobile phones, smart watches, log tables, or any electronic gadgets is strictly prohibited.

Science

1. A beam of light consisting of both paraxial and marginal rays hits a large concave mirror. A screen is placed at the principal focus (f). How will the image on the screen appear, and what is the relation between the focal length for paraxial rays (f_p) and marginal rays (f_m)?
 - (A) Sharp point; $f_p = f_m$
 - (B) Blurred circle; $f_p > f_m$
 - (C) Sharp point; $f_p < f_m$
 - (D) Blurred circle; $f_p < f_m$
2. During the **"Light-Independent Reaction"** (Calvin Cycle), if the supply of NADPH and ATP from the light reaction is suddenly cut off, **which molecule will immediately accumulate in the chloroplast?**
 - (A) Glucose.
 - (B) Oxygen.
 - (C) Phosphoglyceric Acid (PGA).
 - (D) Ribulose Bisphosphate (RuBP).
3. Identify the type of reaction for each of the following as

Combination – (p)
Decomposition – (q)
Displacement – (r)
Double displacement – (s)
Combustion – (t)

(i) A compound breaks apart into its elements
(ii) A metal and a non-metal react to form an ionic compound.
(iii) A compound of Hydrogen and Carbon reacts with Oxygen to produce Carbon- dioxide and water
(iv) Silver ion from AgNO_3 (aq.) forms a precipitate with bromide ion from KBr (aq).

(A) (i) - q (ii) - p (iii) -r (iv) - s
(B) (i) - q (ii) - p (iii) - t (iv) - s
(C) (i) - s (ii) - r (iii) -q (iv) - p
(D) (i) - p (ii) -t (iii) - q (iv) – r

4. The Zinc ions (Zn^{2+}) dissolve into the ocean water. Over time, the Zinc block literally dissolves away atom by atom to provide those protective electrons. This is why the Zinc disappears while the steel hull remains perfectly intact—the Zinc "sacrificed" itself to

prevent the steel from corroding. Which of the following best describes the movement of electrons in this system?

- (A) From the salt water to the Zinc
- (B) From the Steel hull to the Zinc block
- (C) From the Zinc block to the Steel hull
- (D) From the air to the Steel hull

5. A patient's lab report indicates a significantly high level of urea in the blood, while the glucose and protein levels in the urine are normal. Based on the physiology of the nephron, which specific process is likely functioning correctly, and where might the primary pathology lie?

- (A) The Glomerulus is failing to filter large molecules like proteins, causing them to leak.
- (B) Ultrafiltration is functioning normally, but there is a failure in the urea secretion mechanism in the distal tubule.
- (C) Tubular reabsorption of glucose and proteins is efficient, but the overall filtration rate of nitrogenous waste is compromised.
- (D) The Loop of Henle is unable to maintain the osmotic gradient, leading to excess urea retention.

6. An aquarium is filled with a special transparent oil instead of water. A gold coin at the bottom appears to be at a depth of **15 cm**. If the actual depth of the tank is **24 cm**, what will happen to the apparent position of the coin if the oil is replaced by water

(Refractive index of water = 1.33)?

(A) The coin will appear to rise higher than 15 cm.

(B) The coin will appear to sink lower than 15 cm but remain above the bottom.

(C) The coin will appear at the exact same depth of 24 cm.

(D) The coin will appear at its actual depth of 24 cm.

7. Hydrogen gas is not liberated when Cu is reacted with Conc. HNO_3 because:

- (A) Cu is less reactive than hydrogen.
- (B) Cu is more reactive than hydrogen.
- (C) Conc. HNO_3 is strong reducing agent.
- (D) Conc. HNO_3 is strong oxidising agent.

8. A researcher is studying "Regeneration" in Planaria. He cuts one Planaria into three pieces: Head, Middle, and Tail. After two weeks, he has three complete Planaria. Which of the following is the most accurate biological conclusion?

- (A) Planaria use mitosis to increase their size but not to create new tissues.
- (B) Planaria contain "Neoblasts" (specialized stem cells) throughout their body that can differentiate into any missing body part.
- (C) The middle piece only survived because it was already a complete organism.
- (D) Regeneration is a form of sexual reproduction.

9. Four identical resistors, each of resistance R , are connected to form a square. A battery of voltage V is connected across two diagonally

opposite corners of the square. Find
(i) the equivalent resistance of the circuit, and

(ii) the total heat produced in time t .

(A) $R, \frac{V^2}{R} t$ (B) $4R, \frac{V^2}{4R} t$

(C) $\frac{R}{2}, \frac{2V^2}{R} t$ (D) $2R, \frac{V^2}{2R} t$

10. In a laboratory, a human egg cell is fertilized in a petri dish. At the 8-cell stage, one cell is removed for genetic testing. The remaining 7 cells continue to develop into a healthy embryo. What does this experiment demonstrate about early embryonic development?

(A) Each early embryonic cell is "totipotent" and has the potential to develop into a full organism or repair the loss.

(B) Embryos do not need all their cells to survive.

(C) Human reproduction is purely a chemical process with no biological regulation.

(D) The removed cell was a waste product of fertilization.

The Observation Chart

11. Analyze the data in the table below to determine what is happening inside the glow stick:

Feature	Before Snapping	After Snapping
State of Matter	Two separate Liquids	One mixed Liquid
Container Property	Flexible Plastic	Flexible Plastic
Energy Release	None (Dark)	Light Produced (Glowing)

Reversibility	N/A	Cannot be "un-snapped" or turned off
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(A) **Evaporation:** The liquids are being heated and turning into a gas.

(B) **A Chemical Reaction:** The production of light and the fact that it cannot be reversed shows a new substance has been formed, which is different from a simple change of state.

(C) **Transparency:** The plastic tube has changed its property to allow external light to pass through.

(D) **Freezing:** The cold temperature from the snap has turned the liquid into a solid

12. A clip-on lens used in front of a mobile phone camera consists of a convex lens of power +6 D and a concave lens of power -2 D, placed in contact. The mobile phone camera lens has a power of +20 D. An object of height 5 cm is placed at a distance of 25 cm in front of the clip-on lens system. The image formed by the clip-on lens system acts as the object for the mobile camera lens. The final image formed by the mobile camera is:



- (A) Real, inverted, and diminished
(B) Real, inverted, and magnified
(C) Virtual, erect, and magnified
(D) Virtual, erect, and diminished

13. Assertion: Variation is beneficial to the species but not necessarily for the individual.

Reason: Variation allows a population to survive in a changing

(A) Both Assertion and Reason are true, and the Reason is the correct explanation of the Assertion.

(B) Both Assertion and Reason are true, but the Reason is NOT the correct explanation of the Assertion.

(C) Assertion is true, but the Reason is false.

(D) Assertion is false, but the Reason is true.

14. Your recipe calls for 1 teaspoon of **baking powder**, but you only have **baking soda**. Which of the following is the most effective way to ensure your cake rises properly and tastes good?



(A) Use 1 teaspoon of baking soda alone; it is stronger than baking powder and will provide more lift.

(B) Use 1/4 teaspoon of baking soda plus 1/2 teaspoon of an acid (like lemon juice or vinegar).

(C) Use 1 teaspoon of baking soda and double the amount of sugar to mask the metallic taste.

(D) Use 1/4 teaspoon of baking soda and 1/2 teaspoon of cornstarch to stabilize the bubbles.

15. In an experiment, the "Vagus Nerve" (which carries parasympathetic impulses to the heart) is stimulated. What would be the recorded effect on the "Life Process" of circulation, and which part of the brain coordinates this?

(A) Increase in heart rate; coordinated by the Cerebellum.

(B) Decrease in heart rate; coordinated by the Medulla Oblongata.

(C) Increase in blood pressure; coordinated by the Cerebrum.

(D) No change in heart rate; coordinated by the Spinal Cord.

16. A person is sitting in a darkened cinema hall. Suddenly, the movie ends, and the bright overhead lights are switched on. The person experiences a momentary "blinding" sensation before their vision stabilizes. Which of the following best explains the physiological change occurring in the eye?

(A) The ciliary muscles contract to increase the focal length of the eye lens immediately.

(B) The iris contracts the pupil to reduce the amount of light entering, while the rods become temporarily inactive and cones take over.

(C) The vitreous humor changes its refractive index to block excess light from reaching the retina.

(D) The optic nerve stops sending signals to the brain to prevent damage from the sudden intensity.

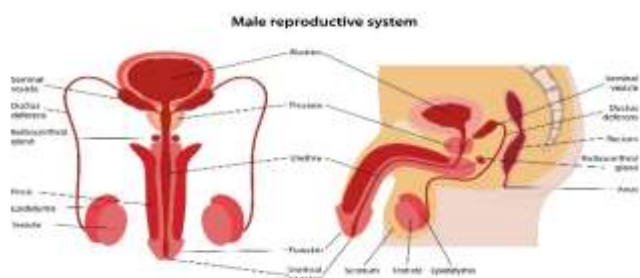
17. In a blast furnace, the temperature in the middle zone accidentally drops by 400°C , causing a significant slowdown in iron production. Which chemical reaction involving carbon monoxide (CO) and iron (III) oxide (Fe_2O_3) is most likely failing due to this temperature drop?

- (A) $\text{Fe}_2\text{O}_3 + 3\text{C} \rightarrow 2\text{Fe} + 3\text{CO}$
- (B) $\text{Fe}_2\text{O}_3 + 3\text{CO} \rightarrow 2\text{Fe} + 3\text{CO}_2$
- (C) $\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$
- (D) $\text{C} + \text{O}_2 \rightarrow \text{CO}_2$

18. An electric kettle rated 2.2 kW, 220 V is used to heat 1 kg of water from 20°C to 100°C . If the efficiency of the kettle is 80%, calculate the time taken and the cost of electricity used if 1 unit (kWh) costs ₹5.

- (Specific heat of water = $4200 \text{ J/kg}^{\circ}\text{C}$)
- (A) 191 s (approximately) and ₹0.58
 - (B) 239 s (approximately) and ₹0.73
 - (C) 150 s (approximately) and ₹0.45
 - (D) 300 s (approximately) and ₹0.90

19. In a diagram showing the "Human Male Reproductive System," which labeled part is responsible for both the production of sperm and the secretion of the hormone Testosterone?



- (A) Prostate Gland
- (B) Vas deferens
- (C) Testis
- (D) Seminal Vesicle

20. While frying vegetables on a gas stove, the bottom of the pan becomes covered with a thick layer of black, powdery soot. What does this indicate about the combustion of the fuel (LPG)?

- (A) Complete combustion of LPG producing carbon dioxide and water
- (B) Incomplete combustion of LPG due to insufficient oxygen
- (C) Presence of excess oxygen causing oxidation of the pan
- (D) Decomposition of the metal of the pan due to high temperature

21. A circular coil of radius **10 cm** consists of **500 turns** of wire. If a current of **2 A** flows through the coil and the magnetic flux density at the center is measured, what would happen to the flux density if the **radius is doubled to 20 cm** and the **current is halved to 1 A**?

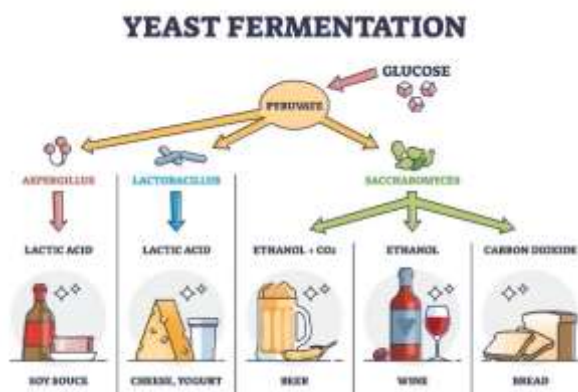
- (A) The flux density remains the same.
- (B) The flux density becomes four times the original.
- (C) The flux density becomes one-fourth of the original.
- (D) The flux density becomes half of the original.

22. Pure alumina (Al_2O_3) has a melting point above 2000°C , making its direct electrolysis commercially impractical. In the Hall-Héroult process, cryolite (Na_3AlF_6) and fluorspar (CaF_2) are added. Why are these additives essential for the commercial viability of the process?

- (A) They increase the melting point of alumina and prevent oxidation of aluminium
- (B) They act as catalysts and are consumed during electrolysis

- (C) They lower the melting point of the mixture and increase its electrical conductivity
- (D) They remove iron impurities and prevent corrosion of the electrolytic cell

23. Refer to the diagram of "Anaerobic Respiration" in yeast. Which gas is represented as a byproduct during the production of Ethanol?



- (A) Oxygen
- (B) Nitrogen
- (C) Carbon Dioxide
- (D) Methane

24. Neutralization Analysis Chart

Component	Hydrochloric Acid (HCl)	Sodium Hydroxide (NaOH)
Role	Acid	Base
State of Matter	Liquid	Liquid
Measured Volume	25 mL	? (Required)
Concentration (M)	0.1m	0.1m
Measurement Unit	Milliliters (mL)	Milliliters (mL)
Instrument Used	Measuring Cylinder/Cup	Measuring Cylinder/Cup

what volume of 0.1 M NaOH is required?

- (A) It would stay at 25 mL because the concentration is the same.
- (B) It would increase to 50 mL because volume is a measurement of space taken up by a liquid.
- (C) It would decrease because the base is "man-made".
- (D) No base would be needed because the acid would evaporate.

25. An electron enters a uniform magnetic field of 0.01 T moving at a speed of 10^6 m/s perpendicular to the field. Simultaneously, a 5 A current is passed through a straight wire placed 2 cm away from the electron's path. If the magnetic field due to the wire and the external magnetic field are in opposite directions at the electron's position, **what is the net force on the electron?**

- (A) 1.6×10^{-15} N (B) 8.0×10^{-16} N
- (C) 2.4×10^{-15} N (D) 1.2×10^{-15} N

26. A person is diagnosed with "Diabetes Mellitus." A doctor finds that their pancreas is producing plenty of Insulin, but their cells are not responding to it. **This "Application-Based" scenario suggests the pathology lies where?**

- (A) In the Alpha cells of the Pancreas.
- (B) In the Insulin receptors on the surface of the body's cells.
- (C) In the person's diet, which lacks glucose.
- (D) In the kidneys, which are producing too much urine.

27. Flameless Ration Heaters (FRHs) contain magnesium powder that reacts vigorously with water to produce magnesium hydroxide ($Mg(OH)_2$) and hydrogen gas (H_2), releasing a large amount of heat. What does this indicate about the nature of the reaction?

- (A) Endothermic, because heat is absorbed from the surroundings
- (B) Exothermic, because heat is released to warm the food
- (C) Endothermic, because hydrogen gas is formed
- (D) Neither endothermic nor exothermic, as no energy change occurs

28. Aromatic compounds were originally named for their distinct odors. Based on your knowledge of materials, which statement best explains the modern scientific view of these substances?

- (A) They are all liquids that take the shape of their container.
- (B) They are all natural materials that come only from animals.
- (C) They are defined by a specific "fixed" circular arrangement of particles (atoms), regardless of their smell.
- (D) They are all man-made materials used to create plastic.

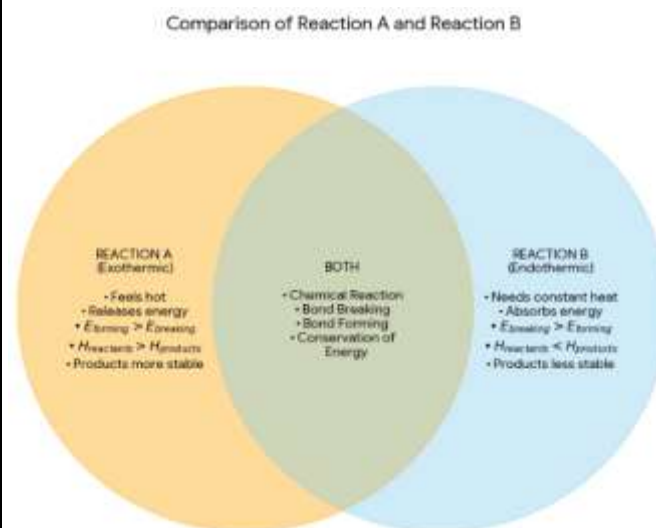
29. A step-up transformer has a turns ratio of 1 : 10. The primary is connected to an AC source of 220 V, and the secondary is connected to a purely resistive load of 22 k Ω . Calculate the power delivered to the load and the current in the primary circuit (assume 100% efficiency).

- (A) 220 W, 1 A (B) 22 W, 0.1 A
- (C) 220 W, 10 A (D) 110 W, 0.5 A

30. A scientist notices that a plant in a sealed jar stops growing after a few days. However, when he adds a small mouse into the jar (with food), both the plant and the mouse survive longer. What does this demonstrate?

- (A) The mouse is eating the plant to survive.
- (B) The plant provides oxygen for the mouse, and the mouse provides carbon dioxide for the plant's photosynthesis.
- (C) The mouse is providing heat to the plant.
- (D) Plants grow better when they have company.

31. A scientist observes two reactions. Reaction A feels hot to the touch, while Reaction B requires constant heating to keep it going.



Which conclusion is scientifically sound?

- (A) Reaction A has a total energy of reactants less than the products.
- (B) Reaction B is more stable because endothermic compounds are always more stable.

(C) In Reaction A, the energy released forming new bonds is greater than the energy used to break old ones.

(D) Reaction B will proceed on its own once it is started.

32. During a 48-hour "Fast" (not eating), which of the following integrated hormonal responses will occur to maintain the "Life Process" of cellular respiration?

(A) The Pancreas secretes Insulin to lower blood sugar.

(B) The Pancreas secretes Glucagon to trigger the Liver to convert stored Glycogen into Glucose.

(C) The Thyroid gland slows down to stop all energy use.

(D) The Adrenal gland secretes Adrenaline to make the person feel full.

33. For the cyanate ion (OCN^-), two resonance structures are drawn:

1. Negative charge on Oxygen ($[\text{O}^- - \text{C} \equiv \text{N} \leftrightarrow \text{O} = \text{C} = \text{N}^-]$)
2. Negative charge on Nitrogen ($[\text{O} = \text{C} = \text{N}^-]$)

Why is the structure with the negative charge on **Oxygen** more stable?

(A) Oxygen is less electronegative than Nitrogen.

(B) Oxygen is more electronegative than Nitrogen and better stabilizes negative charge.

(C) Nitrogen can never carry a negative charge.

(D) Carbon must always be the center of charge.

34. Assertion: Mendel selected pea plants for his experiments because they are cross-pollinating by nature.

Reason: Pea plants show several pairs of contrasting visible characters.

(A) Both Assertion and Reason are true, and the Reason is the correct explanation of the Assertion.

(B) Both Assertion and Reason are true, but the Reason is NOT the correct explanation of the Assertion.

(C) Assertion is true, but the Reason is false.

(D) Assertion is false, but the Reason is true.

35. A concave mirror is kept inside a tank filled with water. A ray of light falls on the mirror. How does the presence of water affect (i) the laws of reflection and (ii) the focal length of the mirror?

Which of the following statements is correct?

(A) The laws of reflection change, and the focal length increases.

(B) The laws of reflection remain unchanged, and the focal length remains the same.

(C) The laws of reflection remain unchanged, and the focal length decreases.

(D) The laws of reflection change, and the focal length remains the same.

36. A pea plant with genotype Tt (Tall) is crossed with another plant of genotype tt (Short). If 500 seeds are collected from this cross and planted, what is the most statistically likely distribution of phenotypes among the resulting plants?

- (A) 500 Tall plants and 0 Short plants.
- (B) 375 Tall plants and 125 Short plants.
- (C) 250 Tall plants and 250 Short plants.
- (D) 125 Tall plants and 375 Short plants.

37. Assertion (A): If an Iron nail is placed in a solution of Magnesium Sulphate, no visible chemical change occurs.

Reason (R): Iron is less reactive than Magnesium in the reactivity series and cannot displace it from its salt solution.

- (A) Both A and R are true and R is the correct explanation of A.
- (B) Both A and R are true but R is NOT the correct explanation of A.
- (C) A is true but R is false.
- (D) A is false but R is true.

38. Which of the following best describes the "Refractory Period" of a neuron, and **why is it essential for signal transmission?**

- (A) It is the time when the neuron is at rest; it allows the brain to sleep.
- (B) It is a brief period after an impulse where the neuron cannot fire again, ensuring signals travel in only one direction.
- (C) It is the moment when neurotransmitters are recycled back into the axon terminal.
- (D) It is the time it takes for a signal to jump from one Node of Ranvier to the next.

39. The correct order of increasing basicity in the following may be:

- (A) $\text{Ba(OH)}_2 < \text{Ca(OH)}_2 < \text{Mg(OH)}_2 < \text{Be(OH)}_2$

- (B) $\text{Be(OH)}_2 < \text{Mg(OH)}_2 < \text{Ca(OH)}_2 < \text{Ba(OH)}_2$

- (C) $\text{Ba(OH)}_2 < \text{Mg(OH)}_2 < \text{Ca(OH)}_2 < \text{Be(OH)}_2$

- (D) $\text{Mg(OH)}_2 < \text{Ca(OH)}_2 < \text{Be(OH)}_2 < \text{Ba(OH)}_2$

40. A point source of light is placed at the bottom of a tank filled with water ($n = 1.33$) to a height of **80 cm**. What is the area of the surface of water through which light from the source can emerge out? (nearly) (CRITICAL ANGLE = 48.6° , $\tan 48.6^\circ \approx 1.13$)

- (A) 2.58 m^2 (B) 6.31 m^2
- (C) 1.45 m^2 (D) 3.14 m^2

41. If the "Progesterone" levels in a pregnant woman's blood were to drop significantly during the second month of pregnancy, **what would be the most likely biological outcome?**

- (A) The fetus would grow faster due to less hormonal interference.
- (B) The uterine lining (endometrium) would break down, likely leading to a miscarriage.
- (C) The mother would begin producing milk immediately.
- (D) The sex of the baby would change from male to female.

42. In a cold country during peak winter, a chemist notices that a bottle labelled "Liquid X" has completely frozen into ice-like crystals, even though the room temperature is around 16°C . When she warms it and adds a few drops of ethanol and concentrated sulfuric acid, a pleasant, fruity smell fills the room.

Why is Liquid X referred to as "Glacial" in many laboratory settings?

- (A) It is harvested from glaciers in cold climates.
- (B) It reacts violently with ice.
- (C) It forms ice-like crystals at temperatures slightly below room temperature.
- (D) It is used to preserve biological samples in cold storage.

43. A patient's blood test reveals that their "Clotting Time" is significantly higher than normal. However, their platelet count is healthy. A genetic analysis shows a deficiency in a specific protein produced by the liver. Based on the "Life Processes" of circulation,

which protein is likely missing?

- (A) Hemoglobin (B) Fibrinogen
- (C) Insulin (D) Pepsinogen

44. Ancient civilizations transitioned from the Copper Age to the Bronze Age by adding tin to copper. **Why is a bronze sword more effective than a pure copper sword?**

- (A) Bronze is lighter than copper, making weapons easier to carry
- (B) Bronze has a lower melting point, making it easier to cast
- (C) Bronze is harder and stronger than pure copper, so it holds an edge better
- (D) Bronze is more abundant in nature than copper.

45. A student uses a convex mirror of focal length 20 cm to view an object. He observes that the linear magnification is 0.5. If he now wants to decrease the magnification to 0.2, in which direction

and by what distance should he move the object?

- (A) 60 cm away from the mirror.
- (B) 20 cm towards the mirror.
- (C) 40 cm away from the mirror.
- (D) 80 cm away from the mirror.

46. Modern laptop chargers often use a three-pin plug, even though the laptop body is mostly plastic. What is the **primary safety function of the third (top) pin** in such an electronic device?

- (A) It acts as a stabilizer to provide a constant voltage to the motherboard.
- (B) It provides a low-resistance path to the ground for any leaked current from the internal metal shielding to prevent electric shock.
- (C) It increases the speed of charging by providing an extra path for the current.
- (D) It is used to filter out high-frequency noise from the AC mains.

47. A person is wearing a corrective lens of power -2.5 D. After a few years, they find they also cannot read a newspaper clearly unless it is held 50 cm away. They decide to switch to bifocals. **What should be the power of the lower part of the bifocal lens?** (Assume standard near point is 25 cm).

- (A) -2.5 D (B) +2.0 D
- (C) +1.5 D (D) +4.5 D

48. A copper wire of diameter **1 mm** carries a current of **1.1 A**. If the number density of free electrons in copper is **$8.5 \times 10^{28} \text{ m}^{-3}$** , calculate the **drift velocity**

and the **current density (J)** in the wire simultaneously. (nearly =)

(A) $v_d = 0.1 \text{ mm s}^{-1}$, $J = 1.4 \times 10^6 \text{ A m}^{-2}$

(B) $v_d = 0.001 \text{ mm s}^{-1}$, $J = 1.4 \times 10^6 \text{ A m}^{-2}$

(C) $v_d = 0.08 \text{ mm s}^{-1}$, $J = 1.1 \times 10^5 \text{ A m}^{-2}$

(D) $v_d = 0.5 \text{ mm s}^{-1}$, $J = 2.5 \times 10^6 \text{ A m}^{-2}$

49. In a genetic cross, a scientist observes that a specific phenotype "skips" a generation—it is present in the grandparents, absent in the parents, but reappears in the grandchildren. **What can be concluded about this trait?**

(A) It is a dominant trait that was suppressed by the environment.

(B) It is a recessive trait, and the parents were likely heterozygous "carriers."

(C) It is an acquired trait that the grandchildren learned from their grandparents.

(D) The trait is caused by a mutation that happens every two generations.

50. A 100 W, 220 V electric bulb and a 60 W, 220 V electric bulb are connected in series across a 220 V power supply. **Which bulb will consume more power in this series arrangement and what is the total power consumed by the circuit?**

(A) 100 W bulb; Total power = 160 W

(B) 60 W bulb; Total power = 37.5 W

(C) 100 W bulb; Total power = 37.5 W

(D) Both bulbs consume equal power; Total power = 80 W

